

# Shadow CSE Website

mode - dark theme

primary-color - #ED7117

why i am starting this website ? and what would it contain ?

1. this website will have a bunch of Computer Science related topics explained pretty well using very easy language. this website is to help students from computer science learn few topics that may not be that well explained in their classes or maybe they have missed those classes or didn't pay attention there. And the goal of this website is to invoke a genuine interest in the students to learn Computer Science by explaining topics with real world examples and telling the application of whatever they are going to learn here in the real world industry out there.
  2. you can get 1:1 live classes as well if free course is not helping you much. or you are fed-up with the tutorials and need some hand holding and personal mentoring for a while.
  3. it will have a bunch of courses like below
    - a. DSA - Data Structures & Algorithms
    - b. DBMS - Database Management System
    - c. Operating System
    - d. System Design
    - e. Computer Networks
    - f. Compiler Design
- design system
    - use the material design system using the shades of this color to build a simple and intuitive UI and the UI should have the contrasts as well using

the multiple tonal shades of these colors like primary, secondary, tertiary, neutral, neutral-variant etc. i have below types of tokens in my system. and i declare these via :root{} declaration. i would like you to keep it consistent with this as well.

```
--md-ref-palette-primary0: #000000;
--md-ref-palette-primary10: #1A1A1A;
--md-ref-palette-primary20: #333333;
--md-ref-palette-primary30: #4D4D4D;
--md-ref-palette-primary40: #666666;
--md-ref-palette-primary50: #808080;
--md-ref-palette-primary60: #999999;
--md-ref-palette-primary70: #B3B3B3;
--md-ref-palette-primary80: #CCCCCC;
--md-ref-palette-primary90: #E6E6E6;
--md-ref-palette-primary95: #F2F2F2;
--md-ref-palette-primary98: #FAFAFA;
--md-ref-palette-primary99: #FDFDFD;
--md-ref-palette-primary100: #FFFFFF;

/* Secondary: Gray */
--md-ref-palette-secondary0: #000000;
--md-ref-palette-secondary10: #1C1C1C;
--md-ref-palette-secondary20: #333333;
--md-ref-palette-secondary30: #4F4F4F;
--md-ref-palette-secondary40: #6B6B6B;
--md-ref-palette-secondary50: #888888;
--md-ref-palette-secondary60: #A3A3A3;
--md-ref-palette-secondary70: #BFBFBF;
--md-ref-palette-secondary80: #D9D9D9;
--md-ref-palette-secondary90: #EBEBEB;
--md-ref-palette-secondary95: #F5F5F5;
--md-ref-palette-secondary98: #FAFAFA;
--md-ref-palette-secondary99: #FDFDFD;
--md-ref-palette-secondary100: #FFFFFF;

/* Tertiary: Lighter Gray */
--md-ref-palette-tertiary0: #000000;
--md-ref-palette-tertiary10: #1E1E1E;
--md-ref-palette-tertiary20: #363636;
--md-ref-palette-tertiary30: #525252;
--md-ref-palette-tertiary40: #707070;
--md-ref-palette-tertiary50: #8C8C8C;
--md-ref-palette-tertiary60: #A8A8A8;
--md-ref-palette-tertiary70: #C4C4C4;
--md-ref-palette-tertiary80: #DDDDDD;
--md-ref-palette-tertiary90: #EEEEEE;
--md-ref-palette-tertiary95: #F7F7F7;
--md-ref-palette-tertiary98: #FBFBFB;
```

```

--md-ref-palette-tertiary99: #FDFDFD;
--md-ref-palette-tertiary100: #FFFFFF;

/* Error: Red (for mistakes) */
--md-ref-palette-error0: #000000;
--md-ref-palette-error10: #410E0B;
--md-ref-palette-error20: #601410;
--md-ref-palette-error30: #8C1D18;
--md-ref-palette-error40: #B3261E;
--md-ref-palette-error50: #DC362E;
--md-ref-palette-error60: #E46962;
--md-ref-palette-error70: #EC928E;
--md-ref-palette-error80: #F2B8B5;
--md-ref-palette-error90: #F9DEDC;
--md-ref-palette-error95: #FCEEEE;
--md-ref-palette-error99: #FFFBF9;
--md-ref-palette-error100: #FFFFFF;

/* Neutral: Pure Black to White */
--md-ref-palette-neutral0: #000000;
--md-ref-palette-neutral5: #0D0D0D;
--md-ref-palette-neutral10: #1A1A1A;
--md-ref-palette-neutral15: #262626;
--md-ref-palette-neutral20: #333333;
--md-ref-palette-neutral25: #404040;
--md-ref-palette-neutral30: #4D4D4D;
--md-ref-palette-neutral35: #595959;
--md-ref-palette-neutral40: #666666;
--md-ref-palette-neutral50: #808080;
--md-ref-palette-neutral60: #999999;
--md-ref-palette-neutral70: #B3B3B3;
--md-ref-palette-neutral80: #CCCCCC;
--md-ref-palette-neutral90: #E6E6E6;
--md-ref-palette-neutral95: #F2F2F2;
--md-ref-palette-neutral98: #FAFAFA;
--md-ref-palette-neutral99: #FDFDFD;
--md-ref-palette-neutral100: #FFFFFF;

/* Neutral Variant: Slightly warmer grays */
--md-ref-palette-neutral-variant0: #000000;
--md-ref-palette-neutral-variant5: #0E0E0E;
--md-ref-palette-neutral-variant10: #1B1B1B;
--md-ref-palette-neutral-variant15: #272727;
--md-ref-palette-neutral-variant20: #343434;
--md-ref-palette-neutral-variant25: #414141;
--md-ref-palette-neutral-variant30: #4E4E4E;
--md-ref-palette-neutral-variant35: #5A5A5A;
--md-ref-palette-neutral-variant40: #676767;
--md-ref-palette-neutral-variant50: #818181;
--md-ref-palette-neutral-variant60: #9A9A9A;
--md-ref-palette-neutral-variant70: #B4B4B4;
--md-ref-palette-neutral-variant80: #CDCDCD;

```

```

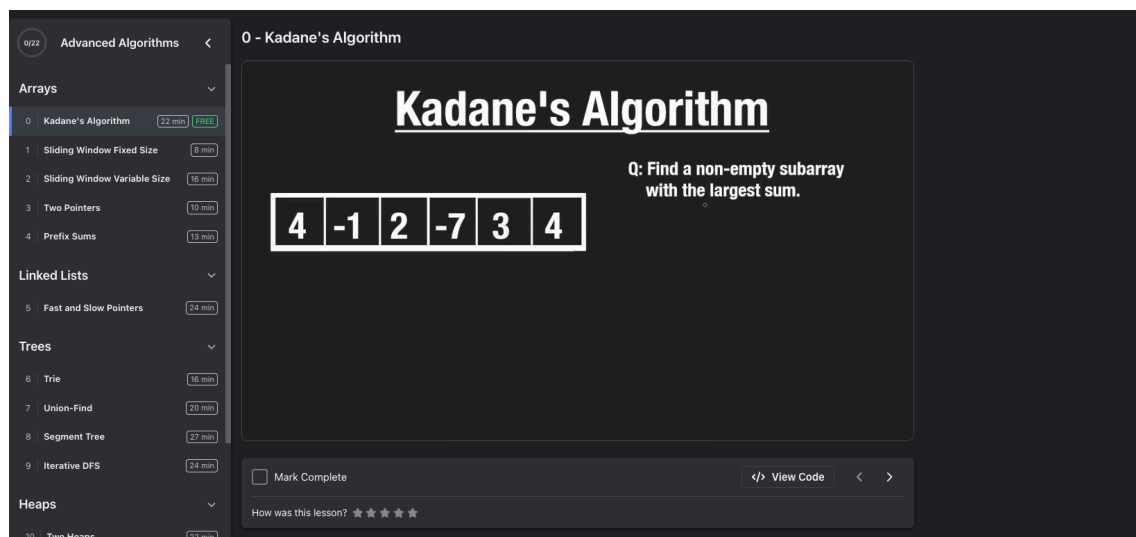
--md-ref-palette-neutral-variant90: #E7E7E7;
--md-ref-palette-neutral-variant95: #F3F3F3;
--md-ref-palette-neutral-variant98: #FAFAFA;
--md-ref-palette-neutral-variant99: #FDFDFD;
--md-ref-palette-neutral-variant100: #FFFFFF;

/* Success: Green (for correct numbers) */
--md-ref-palette-success0: #000000;
--md-ref-palette-success10: #0B2A0F;
--md-ref-palette-success20: #163E18;
--md-ref-palette-success30: #1E5820;
--md-ref-palette-success40: #2B7A2B;
--md-ref-palette-success50: #3D9A3D;
--md-ref-palette-success60: #58B558;
--md-ref-palette-success70: #7BC77B;
--md-ref-palette-success80: #A1D9A1;
--md-ref-palette-success90: #C9EBC9;
--md-ref-palette-success95: #E5F6E5;
--md-ref-palette-success99: #F8FDF8;
--md-ref-palette-success100: #FFFFFF;

```

## ▼ Features

1. a view on left to show the whole topics and subtopics and on right a video player and a place for feedbacks



2. a place to track what sections do we have on this page. a place to attach the list of problems that you should be able to solve based on this video. this will be rendered below the video.

here on the right side, questions and answers should be available as well. so that this space can be used.

☐ Mark Complete
 View Code
<
>

How was this lesson? ★★★★★

### Suggested Problems

| Status                   | Star | Problem                       | Difficulty | Solution |
|--------------------------|------|-------------------------------|------------|----------|
| <input type="checkbox"/> | ★    | Maximum Subarray              | Medium     |          |
| <input type="checkbox"/> | ★    | Maximum Sum Circular Subarray | Medium     |          |

## Kadane's Algorithm

Kadane's algorithm is a greedy/dynamic programming algorithm that can be used on an array. It is used to calculate the maximum sum subarray ending at a particular position and typically runs in  $O(n)$  time.

### Motivation

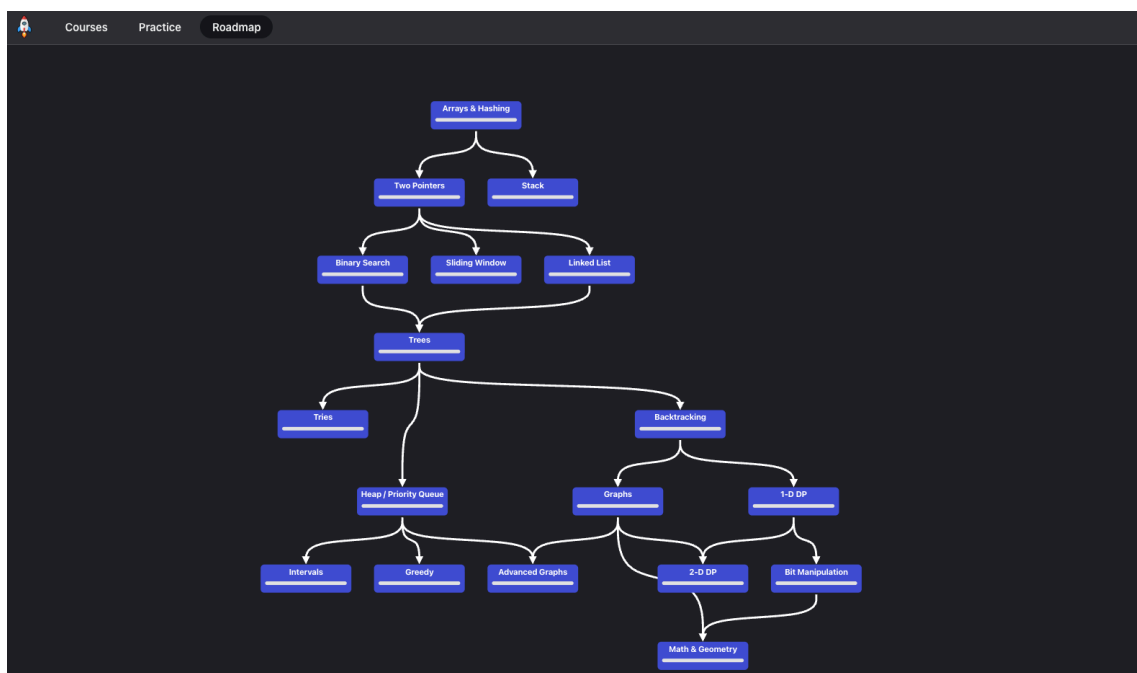
Q: Find a non-empty subarray with the largest sum.

We need to find a group of contiguous elements in an array that result in the maximal sum.

#### Table of Contents

- Motivation
- Optimizing with Kadane's Algorithm
  - Sliding Window
- Time & Space Complexity
  - Time
  - Space
- Closing Notes

- create me a design where i can show what kind of questions the user/student should be able to solve now. kind of like below UI.



#### 4. Home Page

- a. create a home page that will talk about what this website can offer them
- b. use the details inside `why i am starting this website ? and what would it contain ?` section to design the home page of this website. Add some kind of computer science related image to show what this is all about.